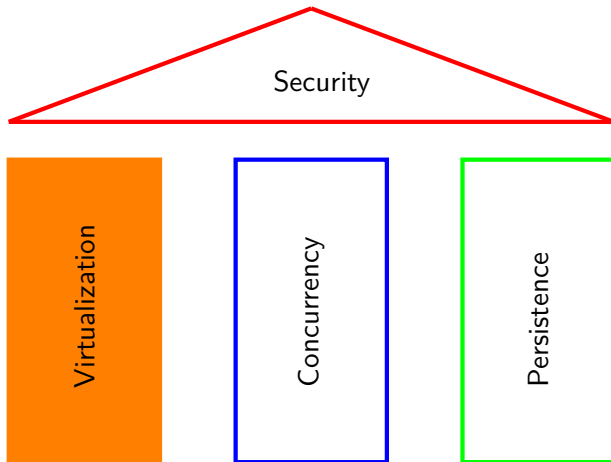


Operating Systems

Virtual CPU (Processes and Threads)

Qiyu Liu

SWU, Spring 2025



Topics covered in this lecture

- The (virtual) process abstraction
- A notion on address spaces
- How processes are created
- Interaction between processes and the OS

What is a Process?



Figure 1: Processes are controlled by the Operating System

Status quo¹:

- CPUs execute an endless stream of instructions (in memory)
- All system memory is in a contiguous physical address space
- The disk is a finite set of blocks
- All instructions execute in privileged mode

To handle concurrent programs, the OS must *separate* the execution of different programs, providing the *illusion* to programs that each program is ***the only running program***.

¹Some simplifying assumptions apply to make our life easier.

Status quo¹:

- CPUs execute an endless stream of instructions (in memory)
- All system memory is in a contiguous physical address space
- The disk is a finite set of blocks
- All instructions execute in privileged mode

To handle concurrent programs, the OS must *separate* the execution of different programs, providing the *illusion* to programs that each program is ***the only running program***.

The ***virtual process abstraction*** provides this illusion.

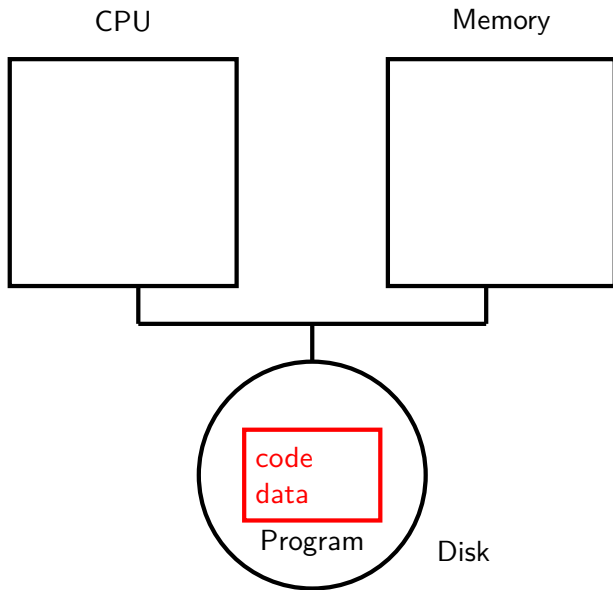
¹Some simplifying assumptions apply to make our life easier.

- A **program** consists of static code and data, e.g., on the disk.
- A **process** is an instance of a program (at any time there may be 0 or more instances of a program running, e.g., a user may run multiple concurrent shells).

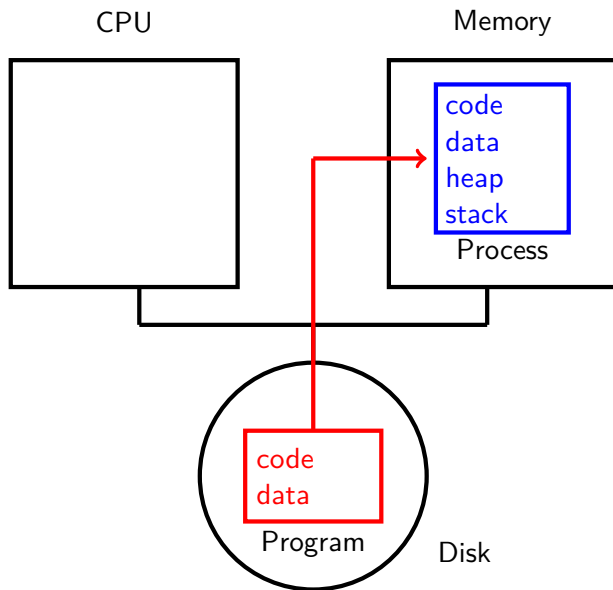
*A **process** is an **execution stream** in the context of a **process state**. The execution stream is the sequence of executing instructions (i.e., the “thread of control”). The process state encompasses everything that executing instructions can affect or are affected by (e.g., registers, address space, persistent state such as files).*

Note: state has two sides, the process view and the OS view. The OS keeps track of the address space and persistence.

Process creation process (1/2)



Process creation process (2/2)

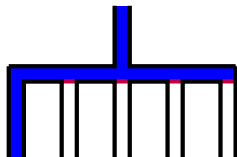


Comparison of terms:

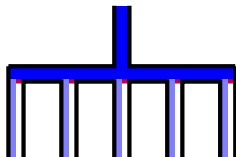
- A **program** is on-disk application, consisting of code and data; programs become a process when they are executed
- A **process** is a running instance of a program. A process starts with a single thread of execution and an address space.
- A process can launch multiple **threads** of execution in the same address space. Each thread receives its own stack but they share global data, code, and heap.

Sharing resources: two forms

Sharing resources: two forms

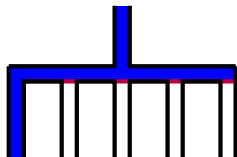


Time sharing (one at a time)

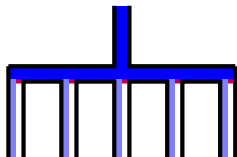


Space sharing (all a little)

Sharing resources: two forms



Time sharing (one at a time)



Space sharing (all a little)

- Shared in time (I get to use the toolbox exclusively)
- Shared in space (I get to pick the two screwdrivers I need)

Virtualizing the CPU

- Goal: give each process the illusion of exclusive CPU access
- Reality: the CPU is a shared resource among all processes
- Two approaches: shared in time or space
 - ***time sharing***: exclusive use, one at a time
 - ***space sharing***: everyone gets a small chunk all the time

Virtualizing the CPU

- Goal: give each process the illusion of exclusive CPU access
- Reality: the CPU is a shared resource among all processes
- Two approaches: shared in time or space
 - **time sharing**: exclusive use, one at a time
 - **space sharing**: everyone gets a small chunk all the time
- Different strategies for CPU, memory, and disk
 - **CPU**: time sharing, alternate between tasks
 - **Memory**: space sharing (more later)
 - **Disk**: space sharing (more later)

OS provides process abstraction

- When the user executes a program, the OS creates a process.
- OS time-shares CPU across multiple processes.
- OS scheduler picks **one** of the executable processes to run.
 - Scheduler must keep a list of processes
 - Scheduler must keep metadata for policy

Difference between policy and mechanism

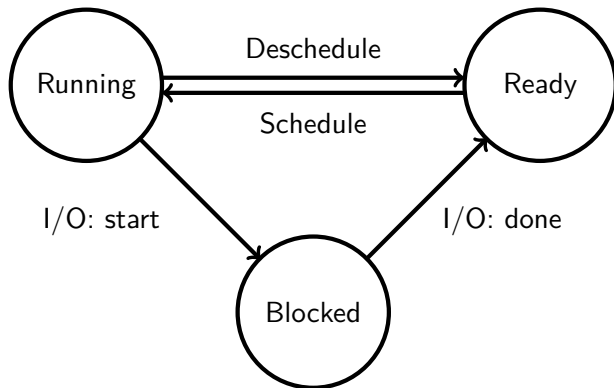
- **Policy:** which process to run
- **Mechanism:** how to switch from one process to another

Distinction between policy and mechanism enables modularity. The scheduling policy is independent of the context switch functionality.

- OS allocates internal data structures
- OS allocates an address space
 - Loads code, data from disk
 - Creates runtime stack, heap
- OS opens basic files (STDIN, STDOUT, STDERR)
- OS initializes CPU registers

- **Running**: this process is currently executing
- **Ready**: this process is ready to execute (and will be scheduled when the policy decides so)
- **Blocked**: this process is suspended (e.g., waiting for some action; OS will unblock it when that action is complete)
- **New**: this process is being created (to ensure it will not be scheduled)
- **Dead**: this process has terminated (e.g., if the parent process has not read out the return value yet)

Process state transitions



Example: process state transitions

Time	Process 0	Process 1	Notes
1	Running	Ready	
2	Running	Ready	
3	Running	Ready	P0 initiates I/O
4	Blocked	Running	P0 is blocked, P1 runs
5	Blocked	Running	
6	Blocked	Running	
7	Blocked	Running	I/O completes
8	Ready	Running	P1 is complete/exits
9	Running	-	

What process should be scheduled if all processes are blocked?

What process should be scheduled if all processes are blocked?

The *idle* process.

Modern kernels use a low priority idle process that is scheduled and executes if no other process is ready. The idle process never blocks or executes any I/O.

What process should be scheduled if all processes are blocked?

The *idle* process.

Modern kernels use a low priority idle process that is scheduled and executes if no other process is ready. The idle process never blocks or executes any I/O.

The idle process is a simple solution to a challenging problem. Without the idle process, the scheduler would have to check if no processes are ready to run and would have to conservatively take action. The idle process guarantees that there is always **at least one** process to run.

- OS maintains data structure (array/list) of active processes.
- Information for each process is stored in a process control block (on Linux, this is called `task struct`) that contains:
 - Process identifier (PID)
 - Process state (e.g., ready)
 - Pointer to parent process (cat `/proc/self/status`)
 - CPU context (if process is not running)
 - Pointer to address space (cat `/proc/self/maps`)
 - Pointer to list of open files (file descriptors, cat `/proc/self/fdinfo/*`)

- **Program:** consists of an executable on disk. Contains all information to bootstrap a process
- **Process:** a running instance of a program; has data section and stack initialized
- **Thread:** a process can have multiple threads in the same address space (computing on the same data)

Distinction between processes and threads

- A thread is a “lightweight process” (LWP)
 - A thread consists of a stack and register state (stack pointer, code pointer, other registers).
 - Each process has one or more threads.

For example, two processes reading address 0xc0f3 may read different values. While two threads in the same process will read the same value.

- Processes can request services through the system call API (Application Programming Interface).
- System calls transfer execution to the OS (the OS generally runs at higher privileges, enabling privileged operations).
- Sensitive operations (e.g., hardware access, raw memory access) require (execution) privileges.
- Some system calls (e.g., read, write) may cause the process to block, allowing the OS to schedule other processes.
- Libraries (the libc) hide system call complexity, export OS functionality as regular function calls.

The process API enables a process to control itself and other processes through a set of system calls:

- `fork()` creates a new child process (a copy of the process)
- `exec()` executes a new program
- `exit()` terminates the current process
- `wait()` blocks the parent until the child terminates
- This is a small subset of the complex process API (more later)

Process API: `fork()`, creating a new process

- The OS allocates data structures for the new process (child).
- The OS makes a copy of the caller's (parent's) address space.
- The child is made ready and added to the list of processes.
- `fork()` returns different values for parent/child.
- Parent and child continue execution in ***their own separate copy*** of their address space (next week: how can we efficiently handle the copy of address spaces?)

Process API: fork() demo!

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char* argv[]) {
    printf("Hello, I'm PID %d (%d, %s)\n", (int)getpid(),
        argc, argv[0]);
    int pid = fork();
    if (pid < 0) exit(-1); // fork failed
    if (pid == 0) {
        printf("o/ I'm PID %d\n", (int)getpid());
    } else {
        printf("\n\o, my child is PID %d\n", pid);
    }
    return 0;
}
```

Process API: `exec()`, executing a (new) program

- Always executing the same program is boring (we would need one massive program with all functionality, e.g., `emacs`).
- `exec()` replaces address space, loads new program from disk.
- Program can pass command line arguments and environment.
- Old address space/state is destroyed except for `STDIN`, `STDOUT`, `STDERR` which are kept, allowing the parent to redirect/rewire child's output!

Why do we need `fork()` and `exec()`?

Assume a user wants to start a different program. For that, the operating system needs to create a new process and create a new address space to load the program.

Why do we need `fork()` and `exec()`?

Assume a user wants to start a different program. For that, the operating system needs to create a new process and create a new address space to load the program.

Let's use divide and conquer:

- `fork()` creates a new process with a copy of this address space
- `exec()` creates a new address space for a program
- `clone()` adds a thread (of execution) to this address space

Process API: `wait()`, waiting for a child

- Child processes are tied to their parent.
- `exit(int retval)` takes a return value argument.
- Parent can `wait()` for termination of child and read child's return value.

A tree of processes

- Each process has a parent process
- A process can have many child process
- Each process again can have child processes

```
3621  ?      Ss    \_  tmux
3645  pts/2   Ss+   |    \_  -zsh
3673  pts/3   Ss+   |    \_  -zsh
4455  pts/4   Ss+   |    \_  -zsh
27124 pts/1     Ss+   |    \_  -zsh
21093 pts/5     Ss    |    \_  -zsh
10589 pts/5     T     |    |    \_  vim 02-review.md
10882 pts/5     R+   |    |    \_  ps -auxwf
10883 pts/5     S+   |    |    \_  less
21264 pts/7     Ss    |    \_  -zsh
 1382 pts/7     T     |    |    \_  vim /home/gannimo/notes.txt
14368 pts/9     Ss    |    \_  -zsh
29963 pts/9     S+   |          \_  python
```

A process executes instructions directly on the CPU.

A process executes instructions directly on the CPU.

Issues with running directly on hardware:

- Process could do something illegal (read/write to memory that does not belong to the process, access hardware directly)
- Process could run forever (OS must stay in control)
- Process could do something slow, e.g., I/O (OS may want to switch to another process)

A process executes instructions directly on the CPU.

Issues with running directly on hardware:

- Process could do something illegal (read/write to memory that does not belong to the process, access hardware directly)
- Process could run forever (OS must stay in control)
- Process could do something slow, e.g., I/O (OS may want to switch to another process)

Solution: OS maintains some control with help from hardware. For example, the OS maintains timers to intercept the execution at regular intervals and the process may not execute privileged instructions that access the hardware directly.

Process isolation policy

- On most operating systems, processes are:
 - Isolated from each other
 - Isolated from the OS
- Isolation is a core requirement for security:
 - Constrains bugs to the process
 - Enables privilege isolation
 - Enables compartmentalization (breaking complex systems into independent fault domains)

What *mechanism* allows process isolation?

Process isolation mechanism

- Virtual memory: one (virtual) address space per process
- Different execution modes: OS executes at higher privileges
 - Process executes in user mode (ring 3 on x86)
 - OS executes in super mode (ring 0 on x86)

- Processes are a purely **virtual concept**
- Separating policies and mechanisms enables modularity
- OS is a server, **reacts to requests** from hardware and processes
- Processes are **isolated** from the OS/other processes
 - Processes have no direct access to devices
 - Processes run in virtual memory
 - OS provides functionality through system calls
- A process consists of an address space, associated kernel state (e.g., open files, network channels), and one or more threads of execution

Don't forget the Moodle quiz!